WACE Mathematics Specialist · Cheat Sheet (Patterns + Key Notes)

General rules: Principal argument $-\pi < \theta \le \pi$ · Write mathematical statements, not calculator language · When explaining, do not write "it"; state which quantity (e.g. "SD of the sample mean") · ln|...| include absolute value (may omit when the argument is always positive) · Indefinite integral + c · Rates of change with units · Expand brackets and guard against sign errors when copying (repeatedly flagged in yearly reports) · Memorise exact trigonometric values for special angles · When a question asks to prove / show, use algebra; "CAS says no solution" is not a proof · State the conclusion clearly

1 Complex numbers

① Form conversion and operations

- Division: multiply numerator and denominator by the conjugate of the denominator. Powers and roots: first write in r cis θ form, then use De Moivre ($(\sqrt{3} - i)^5 = 32\,\mathrm{cis}(-5\pi/6) = -16\sqrt{3} - 16i$).
- ⚠ Third quadrant: the principal argument of $-1 - i$ is $-3\pi/4$, not $\pi/4$, and cannot be written as $5\pi/4$. Draw a diagram first to determine the quadrant.
- Given $z = 3 + bi$ and $\tan\theta = 2 \Rightarrow b = 6$, $r = 3\sqrt{5}$.

② Solve $z^n = w$ ★ appears every year

- Write w as R cis φ (1 mark for modulus, 1 mark for argument) → z = R^(1/n) cis((φ + 2kπ)/n), k = 0, ..., n − 1 (1 mark for the general formula) → list all n roots within the range specified in the question, with adjacent roots separated by 2π/n.
- Example: $z^4 = -16i \Rightarrow 2\,\mathrm{cis}(-5\pi/8)$, $2\,\mathrm{cis}(-\pi/8)$, $2\,\mathrm{cis}(3\pi/8)$, $2\,\mathrm{cis}(7\pi/8)$.
- Roots of $z^n = 1$: equally spaced on the unit circle (2 marks for explanation: radius 1, equal spacing); product of all roots = $(-1)^{n+1}$; $z^{43} = 1$ and $z^{47} = 1$ have only the common root 1 (43 and 47 are coprime).
- z^n is a positive real number: n·arg z is an integer multiple of 2π ($z = 4\,\mathrm{cis}(5\pi/6) \Rightarrow$ smallest n = 12).

③ Complex roots of polynomials

- "Show (z − α) is a factor": calculate P(α), write out every term, obtain 0 (key: not just the statement).
- Real coefficients ⇒ $\bar{\alpha}$ is also a root ⇒ $(z - \alpha)(z - \bar{\alpha}) = z^2 - 2\mathrm{Re}(\alpha)z + |\alpha|^2$ → use division or compare coefficients to find the remaining factor → solve the quadratic.
- Example: $P(2 + 4i) = 0 \Rightarrow$ factor $z^2 - 4z + 20$, remaining $z^2 - 2z + 3 \Rightarrow z = 1 \pm \sqrt{2}\,i$. ⚠ $z^2 - 2z + 3$ cannot be factorised as $(z - 3)(z + 1)$ (2021 report).
- Given two complex roots, find the coefficients: each pair of conjugate roots gives a quadratic factor, then use the constant term to find the real root.

④ Geometric meaning of multiplication

- Template (1 mark each): "Multiplication by $w = r\,\mathrm{cis}\,\theta$ dilates the modulus by a factor r and rotates anticlockwise by θ (clockwise by |θ| if θ < 0) about the origin." Multiplication by z^{-1}: take the reciprocal of the factor and reverse the direction of rotation.
- Find n for which zw^n lies in the second quadrant: each step rotates by arg w; when |w| = 1 there is a period, so give the full set of answers, e.g. n = 3 + 12k, 4 + 12k, 5 + 12k.

⑤ Loci: sketch from equations

- $|z - a| = r$: circle. $|z - a| = |z - b|$: perpendicular bisector of a and b. $\mathrm{Arg}(z - a) = \theta$: ray, with open endpoint at a.
- $|z - a| + |z - b| = k$: an ellipse when k is greater than |a − b|, degenerates to a line segment when equal.
- $|z + 2| = |z - i| + \sqrt{5}$ (difference equals the distance between the two points): ray starting from i, with solid endpoint at i.
- $(z + i)\cdot\mathrm{conj}(z + i) = 2 \Rightarrow$ centre −i, radius $\sqrt{2}$; $z - \bar{z} \le 6i \Rightarrow \mathrm{Im}\,z \le 3$.
- Marking points for graphs: key points or centre, position of boundaries, solid or dashed lines, shaded regions.

⑥ Loci: write equations from graphs

use only z, not x, y; circle ⇒ $|z - c| \le r$; sector ⇒ two Arg inequalities + one modulus inequality; find angles exactly using trigonometry.

⑦ Extrema on loci

- Distance from a point on a circle to c: maximum d + r, minimum |d − r| (d = distance from the centre to c). Nearest point on a ray: draw a perpendicular.
- Maximum argument occurs at the tangent: first use $\sin\theta = r/d$ to find the angle between the tangent and the line to the centre, then add the argument of the centre, noting the range specified in the question.

⑧ Vectors and geometric methods

- z + r (sum of two equal-length vectors) forms a rhombus ⇒ $\arg(z + r) = \theta/2$; z − k forms a triangle, use the sine or cosine rule.
- Regular hexagon: first check where the centre is. When centred at the origin, all vertices have equal modulus and arguments differ by π/3; in 2025F-Q7, z_0 is at the origin, the moduli are r, √3r, 2r, √3r, r in order, and the product is $6r^5$.
- Trick: $\Sigma\, n\cdot i^n$ groups into sets of 4 terms, each group sums to 2 − 2i; taking the modulus of $(a + bi)^3 \Rightarrow (a^2 + b^2)^3 = |...|^2$.

2 Functions and graphs

① Domain of composite functions ★ most frequently flagged (6 years)

- f(g(x)) is defined ⇔ x ∈ dom g and g(x) ∈ dom f. After substituting, do not simplify first; each restriction usually earns 1 mark.
- Sources of restrictions: denominator ≠ 0; expression under a square root ≥ 0; when the square root is in the denominator, > 0; argument of ln > 0.
- Example: $f = 1 - \sqrt{(x - 4)}$, $g = 1/x^2 \Rightarrow 1/x^2 \ge 4 \Rightarrow -\frac{1}{2} \le x \le \frac{1}{2}$ and $x \ne 0$. $f = \sqrt{(4 - x)}$, $g = 1/x^2 \Rightarrow x \le -\frac{1}{2}$ or $x \ge \frac{1}{2}$.
- ⚠ $\sqrt{(x^2)} = |x|$, so f(h(g(x))) is not necessarily equal to f(x).

② Inverse functions

- Exists ⇔ one-to-one (horizontal line test); otherwise restrict the domain, with the "largest possible domain" taken up to the vertex.
- Method: swap x and y (1 mark) → solve for y → the ± sign is determined by ran f^{-1} = dom f; dom f^{-1} = ran f.
- Graph is symmetric about y = x, mark the corresponding endpoints (a, b) ↔ (b, a). Whether $f^{-1}(-1)$ is defined: first check whether −1 is in ran f.
- Inverse trigonometric functions: $x = 2\tan y \Rightarrow y' = 2/(4 + x^2)$ (use implicit differentiation, then use $1 + \tan^2 y$ to convert to x).
- Explain "g is not one-to-one": "because g(1) = g(−1) =

1"; do not write "it is many to one".

③ Sketching y = 1/f(x) ★ often examined in CF

- Zeros of f ⇒ vertical asymptotes; as x → ±∞, f → ±∞ ⇒ horizontal asymptote y = 0.
- Vertical asymptote of f at x = b ⇒ 1/f has an open point at (b, 0) (range excludes 0); horizontal asymptote of f at y = c (c ≠ 0) ⇒ y = 1/c.
- Local maximum (a, m) ↔ local minimum (a, 1/m); points where f = ±1 are fixed; sign does not change; open point of f at (a, b) ⇒ open point of 1/f at (a, 1/b), when b = 0 it becomes a one-sided asymptote.
- Find the range: read from the graph of 1/f, noting values that are not attained.

④ Absolute value functions

- |f(x)|: reflect the part below the x-axis above the x-axis; f(|x|): keep the right half and mirror it to the left; f(−|x|): keep the left half and mirror it to the right.
- Number of solutions to |f(x)| = k: intersect with the horizontal line y = k, check the endpoint k separately; you can sketch the number-of-solutions function N(k).
- The solution set of |x + 1| + |x + a| = k is an interval ⇒ the flat section of the sum of two absolute values is the solution set (a = 3, k = 2).
- Determine y = a|x − b| + c from the graph: the vertex gives b and c, the gradient gives a.

⑤ Rational functions: determine parameters from graphs / sketch graphs

- Zeros ↔ factors of the numerator; vertical asymptotes ↔ factors of the denominator; horizontal asymptote = ratio of leading coefficients; oblique asymptote = quotient from polynomial division (x − 1 − 3/(x + 1) ⇒ y = x − 1).
- Repeated factor: in the numerator ⇒ graph touches but does not cross the x-axis; in the denominator ⇒ both sides of the asymptote have the same sign (both tend to −∞).
- Give a reason (1 mark): "c = 3 since x = 3 is a vertical asymptote". A quadratic with no zeros, a(x − 3)² + c, requires a and c to have the same sign.
- Sketch checklist (1 mark each): x-intercept, y-intercept, vertical asymptote, horizontal or oblique asymptote, turning points (mark if present), behaviour on each side of asymptotes. First check for any cancelled factor (hole).

3 Three-dimensional vectors and systems of linear equations

① Systems of linear equations ★ appeared in CF in 7 of the last 10 years

- Three elimination steps (1 mark each): use two pairs of equations to eliminate the same variable → solve for one variable → back-substitute.
- With parameters: reduce to (k − 2)z = c − 5 ⇒ k ≠ 2: unique solution (c arbitrary); k = 2 and c ≠ 5: no solution; k = 2 and c = 5: infinitely many solutions.
- ⚠ For infinitely many solutions, write the solution set (2022 report: many stopped after writing "infinitely many solutions"), express it using a parameter or relationships between variables. Example: z = t ⇒ y = t − 1, x = (11 − 3t)/2, i.e. r = (11/2, −1, 0) + t(−3/2, 1, 1).
- Geometric interpretation: unique solution = three planes intersect at a point; one parameter = intersect in a line (possibly two planes coincide; both points must be written); two parameters = three planes coincide; no solution = there are parallel planes, or the three planes intersect pairwise to form three parallel lines (triangular prism, 2020F-Q4). First check whether the normal vectors are proportional.

② Plane equations

- Parametric form r = a + λu + μv ⇒ n = u × v (write each component of the cross product clearly) ⇒ r·n = a·n, i.e. ax + by + cz = k.
- Through three points: cross product of two edge vectors; z = 2x + y + 4 ⇒ n = (2, 1, −1).
- Contains line L and is perpendicular to plane 1 ⇒ n_2 = d_L × n_1; contains two intersecting lines ⇒ n = d_1 × d_2.

③ Intersections, angles, distances

- Line and plane: substitute r = a + λd, solve for λ, write the point (3 marks). Line and sphere: obtain a quadratic equation in λ, two solutions, tangent, no intersection; or compare the distance from the centre of the sphere to the line with the radius.
- Two lines: set components equal, use two equations to solve for λ, μ, substitute into the third to check; if it does not hold, they do not intersect, and this step is the proof.
- Closest point on a plane to point L: draw a line through L along the normal vector, find its intersection with the plane.
- Angle between line and plane: sin θ = |d·n|/(|d||n|). Distance from point to plane = |n·(P − A)|/|n|, where A is any point on the plane; this formula is not on the formula sheet, so if used, include the derivation or use the normal method instead. Volume of parallelepiped = |c·(a × b)|.

④ Spheres: complete the square on the general form, e.g. $x^2 + y^2 + z^2 - 4x + 2y - 6z + 5 = 0 \Rightarrow |r - (2, -1, 3)| = 3$; endpoints of a diameter ⇒ centre is the midpoint (report: often calculated incorrectly); tangent to a plane ⇒ radius = distance from centre to plane; passes through all vertices of a rectangular prism ⇒ centre is the midpoint of a space diagonal.

⑤ Vector proofs (in both CF and CA): express the required vector in terms of basis vectors → expand the dot product (key: get 4 terms) → use given conditions (square ⇒ a·b = 0, |a| = |b|) → state conclusion. Marks for notation.

⑥ Applications: reflection ⇒ the sum of the unit vectors B→S and B→R is ∥ n; the steepest descent direction on a slope lies in the slope plane and is perpendicular to the horizontal line; for distance problems involving moving objects, see the next section.

4 Vector calculus and motion (only in CA paper)

① Differentiation: v = r′, a = v′, differentiate component by component; speed |v|; draw vectors on the diagram (with the start point at the position at that time). Constant speed ≠ uniform velocity (direction is changing); write that $\sin^2 + \cos^2 = 1$ was used (1 mark).

② Integrate to find r(t)

- Integrate component by component, add a constant vector, use r(0) to determine each component (2018 report: many wrote this unclearly).
- Gravity + wind: a = (0, −9.8) ⇒ r = ((70cos θ + w)t, 70 sin θ·t − $4.9t^2$); range: set y = 0, solve for t, then substitute into x.
- Horizontal drag + vertical lift: x′ = 32e^(−0.05t), h″ = s

− 9.8, find s from the given r. For the landing angle, use the angle between the velocity vector and the slope.

③ Eliminate the parameter to get a Cartesian equation ★

- $\sin^2 + \cos^2 = 1$ ⇒ circle or ellipse; double-angle formulas directly give the relationship between x and y ⇒ a segment of a parabola; first convert to the same angle and then use $\sin^2 + \cos^2$ ⇒ general curve (e.g. a figure "8" shape); $(e^t + e^{-t})^2 - (e^t - e^{-t})^2 = 4 \Rightarrow x^2 - y^2 = 4$; x is linear in t ⇒ solve directly for t and substitute into y.
- ⚠ Write the range of values, derive it from the range of t; often both x and y need restrictions: $x \geq 2$ and $y \geq 0$ (writing only $x \geq 2$ loses marks); $1 \leq x \leq e^3$.

④ Distance travelled and displacement

- Distance travelled = $\int_a^b \sqrt{(\dot{x}^2 + \dot{y}^2)}\, dt$; first determine the start and end times: time for one lap = least common multiple of the periods of the components (2π, 4π, 6π have all been tested), or find t from the coordinates.
- Explanation: $\int_0^1 v\, dt$ = "the change in displacement (position vector) during the first second"; $\int |v|\, dt$ = distance travelled. The 2025 report specifically names confusion between these two. Draw the displacement vector on the diagram: from r(a) to r(b).

⑤ Closest distance and relative position

- Same time parameter: relative position $d(t) = r_P - r_Q$ (when asked for relative position, the order must be correct, 1 mark); closest ⇔ $d(t) \cdot (v_P - v_Q) = 0$; "how long within 50 m" ⇒ solve $|d(t)| < 50$.
- Both speed and departure time can be adjusted ⇒ shortest distance between two lines: use independent parameters λ, μ, make PQ perpendicular to both d_1 and d_2.
- Angle of descent = $|90° - \varphi|$ (φ is the angle between the velocity and k), or use $\sin^{-1}(|v_z|/|v|)$ directly.

5 Integration techniques and applications

① Definite integrals with a given substitution ★ almost every year in CF (4–7 marks)

- Five steps (1 mark each): find dx from the substitution → change the limits → convert the integrand completely to the new variable and simplify → integrate → substitute to get the exact value.
- Trigonometric substitution: for $\sqrt{(a^2 - x^2)}$ use $x = a \sin\theta$; for $x^2 + a^2$ use $x = a \tan\theta$. Example: $\int_0^{\sqrt{3}} \sqrt{(1 - x^2/4)}\, dx$, $x = 2\sin\theta \Rightarrow \int_0^{(\pi/3)} 2\cos^2\theta\, d\theta = \pi/3 + \sqrt{3}/4$.
- Indefinite integral version ($\int x(1+x)^n\, dx$): do not change the limits; substitute back to x at the end. ⚠ Simplifying after substitution is often where errors occur (2020, 2023 reports).

② Trigonometric identities: $\cos^2 = \frac{1}{2}(1 + \cos 2x)$; $\sin^2 = \frac{1}{2}(1 - \cos 2x)$; $(\sin x + \cos x)^2 = 1 + \sin 2x$; product-to-sum $2\sin A \cos B = \sin(A + B) + \sin(A - B)$; $\tan^2 = \sec^2 - 1$.

③ Partial fractions ★ high frequency in CF

- Forms: distinct linear factors $A/(x - a) + B/(x - b)$; repeated root $A/(x + 1) + B/(x + 1)^2$; irreducible quadratic $(Bx + C)/(x^2 + 2)$. When the question specifies the form, write it as given.
- After putting over a common denominator, equate numerators (1 mark) → substitute special values or compare coefficients to find the constants (1 mark) → integrate term by term: $\ln|x - a|$, $-B/(x + 1)$, if the numerator is the derivative of the denominator then $\ln|$quadratic$|$, $q/(x^2 + 4) \rightarrow (q/2)\tan^{-1}(x/2)$ → write + c.
- Example: $1/(x^2 - 1) \Rightarrow \frac{1}{2} \ln|(x - 1)/(x + 1)| + c$; $(5x + 3)/(x + 1)^2 = 5/(x + 1) - 2/(x + 1)^2$. If no hint is given, first factorise the denominator yourself (1 mark).

④ Area

- First find intersection points and tangent points; when writing the curve as $x = g(y)$ is more convenient, integrate with respect to y (key: marks are given for both methods).
- Symmetry: for an ellipse, or a shape symmetric about a coordinate axis, multiply by a factor; circular segment: difference between the upper and lower branches $2\sqrt{(\ldots)}$.
- A definite integral can be interpreted as a geometric area (triangle, trapezium). Compare the numerical area with πr^2 to determine whether a curve is a circle.

⑤ Volumes of solids of revolution ★ high frequency in CA

- About the x-axis $\pi\int y^2\, dx$; about the y-axis $\pi\int x^2\, dy$ (first write x^2 as a function of y); between two curves $\pi\int(R^2 - r^2)$, ⚠ not $\pi\int(R - r)^2$.
- The integral expression must include π, the limits, and dx or dy (limits + notation together give 1 mark).
- Contexts: depth when 80% full ⇒ set $\int_0^h \pi x^2\, dy = 0.8V$ and solve for h; filling rate $dV/dt = (dV/dh)(dh/dt)$; increment $\delta h \approx \delta V \div \pi x^2|_{(y=h)}$; annulus R, $r = 3 \pm \sqrt{(4 - y^2)}$.

6 Rates of Change and Differential Equations

① Implicit differentiation

- Differentiate both sides with respect to x; terms containing y are multiplied by y′; use the usual product and chain rules. When the question says not to solve for y′, don't solve it—substitute the point directly to evaluate.
- Horizontal tangent: set numerator = 0 and solve simultaneously with the original equation; second derivative: differentiate the expression already found again, then substitute x, y, y′.

② Related rates ★

- Write the relation (eliminate redundant variables if necessary; you can also keep two variables and differentiate implicitly) → differentiate with respect to t → substitute the values at that instant (often find θ, h first) → include units.
- Example: regular hexagon $A = (3\sqrt{3}/2)s^2$, $s = 4$, $ds/dt = 0.5 \Rightarrow dA/dt = 6\sqrt{3} \approx 10.39\ cm^2/s$. Lighthouse makes 3 revolutions per minute ⇒ $d\theta/dt = \pi/10$ rad/s.
- Circular motion (Ferris wheel, merry-go-round): $d\theta/dt = 2\pi/T$; use the cosine rule $s^2 = 89 - 80\cos\theta$ or $y = R\sin(\theta + \alpha)$; differentiate implicitly with respect to t; find the maximum of ds/dt by differentiating again (geometrically when PC is tangent to the circle). ⚠ Height above ground = height relative to the centre + R.

③ Slope fields

- Substitute points to find slopes (1 mark); infer the equation from the field: see whether the slope changes only with x or only with y, where it is 0, and where it is undefined.
- Draw solution curves: pass through the given point, follow the short line segments, mark symmetry; where denominator = 0 and numerator ≠ 0, the tangent is vertical and the curve turns back there; where slope = 0

there is a horizontal tangent; the point of maximum slope is a point of inflection (2025A-Q16 point of inflection at (3, −1)).

④ Separation of variables

- Write as $\int g(y)dy = \int f(x)dx$ (1 mark) → integrate, add only one constant, put absolute value signs around the ln argument → substitute the initial condition → write in explicit form.
- $dP/dt = kP \Rightarrow P = P_0e^{kt}$. ⚠ "instantaneous growth rate is 10%" is $dP/dt = 0.1P$ ($P_0 = 30000e^{(-4.5)} \approx 333$); it cannot be treated as 1.1 times per year (≈ 412).
- Example: $dh/dt = -1/(100h) \Rightarrow h^2 = h_0^2 - t/50$; $dv/dt = -9.8 - 2v \Rightarrow v = 4.9(e^{(-2t)} - 1)$; depth must then be found by integrating v again.

⑤ Logistic growth ★

- $dP/dt = rP(k - P) \Leftrightarrow P = k/(1 + Ae^{(-rkt)})$: carrying capacity = the non-zero P that makes $dP/dt = 0$; fastest growth occurs when $P = k/2$, maximum growth rate = $r(k/2)^2$.
- ⚠ Letters depend on the question (in some questions k is the proportionality constant; in others it is written as $(1/r)P(k - P)$). Example: $P = 18000/(10.25e^{(-0.15t)} + 1) \Rightarrow$ carrying capacity 18000, maximum growth rate 0.15 × 9000 × ½ = 675 horses/year.
- Sketching: $k/2 < P_0 < k \Rightarrow$ concave down throughout, no point of inflection; $P_0 > k \Rightarrow$ decreasing towards k; changing the carrying capacity ⇒ new point of inflection and asymptote.
- Explanation: when $P < k/2$, the growth rate increases with P; when $P > k/2$, it decreases and approaches k. For equations that are not logistic (e.g. $0.001(100 - P)^2$), analyse the change in slope directly.

⑥ Velocity as a function of displacement v = f(x) ★★ flagged in reports three times

- $a = dv/dt = v \cdot dv/dx = d(\frac{1}{2}v^2)/dx$. ⚠ dv/dx itself is not acceleration. Example: $v = -0.2x \Rightarrow a = 0.04x$; $v = x/8 \Rightarrow a = x/64$.
- x(t): $dx/dt = v(x)$, separate variables, substitute the initial position. Example: $v = x/8$, $x(0) = 192 \Rightarrow x = 192e^{(t/8)}$.
- Test for SHM: only $a = -n^2(x - c)$ (proportional to displacement and in the opposite direction) qualifies; $a = 0.04x$ is not.

⑦ Simple harmonic motion (SHM)

- $\ddot{x} = -n^2x \Rightarrow x = A\cos(nt + \varphi)$; $T = 2\pi/n$; $v_{max} = An$; $v^2 = n^2(A^2 - x^2)$.
- Determine A and the phase: $x(0) = 0$ and $v(0) > 0 \Rightarrow A\sin(nt)$; if no direction is given, both $\pm A\sin$ are accepted; $n = 2$, $x(0) = -1$, $v(0)^2 = 32 \Rightarrow A = 3$.
- Distance travelled: 4A per period, 2A per half-period; for less than half a period and when the starting point is not at equilibrium or an endpoint, use $\int|v|\,dt$. Damping: $A = A_0e^{(-kt)}$; treat $A < 0.01$ as stopped.

⑧ Small increments formula: $\delta y \approx (dy/dx)\delta x$; $\delta N \approx (dN/dt)\cdot\delta t$ (dN/dt is given directly by the equation; watch the time units).

7 Statistical Inference (CA paper only)

① Distribution and probability of $\bar{X}$ ★

- "By the CLT, since n is large, $\bar{X}$ is approximately normal with mean μ and SD $\sigma/\sqrt{n}$" (normal, mean, standard deviation worth 3 marks in total). If the population is normal, this holds for any n.
- If the population is not normal (uniform, exponential, skewed): the method is the same; when asked "does the answer change?", answer no + CLT reason.
- Sum question: 50 times with a total < 8.96 kL ⇔ $\bar{X} <$ 179.2 L. Draw the distribution of $\bar{X}$: a narrow bell shape centred at μ, width about $\pm 3\sigma/\sqrt{n}$; shade the probability region when required.

② Finding the sample size

- First draw a diagram to decide whether it is two-tailed, one-tailed or a half-interval, then find the critical z: $P(|\bar{X} - \mu| < d) = 0.98 \Rightarrow z = 2.326$; $P(\bar{X} > 25) = 0.03 \Rightarrow z = 1.881$; $P(0 < Z < k) = 0.4 \Rightarrow k = 1.282$.
- $n = (z\sigma/d)^2$: when "at least" is required, round up (304.3 ⇒ 305, 600.25 ⇒ 601); when finding n from an interval, take the nearest integer (differences caused by using $z = 2.576$ and $z = 2.58$ are both accepted in the key).

③ Confidence intervals

- $\bar{x} \pm z \cdot s/\sqrt{n}$. Reverse-engineer: midpoint = $\bar{x}$; half-width = $z \cdot \sigma_{\bar{X}}$ ($150 \le \mu \le 200 \Rightarrow \sigma_{\bar{X}} = 25/1.96 = 12.76$); find s from the width and n.
- Changing sample size: width ∝ $1/\sqrt{n}$ (n × 16 ⇒ width becomes 1/4); 2n ⇒ standard error divided by √2.

④ Interpreting confidence intervals ★ must memorise (state True / False first, then give a reason)

- μ is fixed but unknown; a particular interval either contains μ or does not, and you cannot know which.
- 95% means that in repeated sampling about 95% of intervals contain μ, not "the probability that μ lies in this interval is 95%".
- The next sample mean does not necessarily lie in the interval; a single individual even less so (variation between individuals is much greater than variation in sample means).
- A wider interval does not mean it is more likely to contain μ; 10 out of 50 intervals being too low is normal variation; but a large number of intervals not containing μ (far more than 5%, e.g. 298 out of 500) is evidence, although it still does not prove the cause.
- Copying the same sample to make 2n is not a new random sample, and the interval does not really become narrower; combining two independent random samples is acceptable and gives greater precision.
- Intervals from different sample sizes cannot be compared directly in this way; a lower sample mean alone does not show the new method is better.

⑤ Using an interval to test a claim: construct the μ interval for the new sample (e.g. $182.5 \pm 1.96 \times 15/6 = (177.6, 187.4)$) → is the claimed value 175 in the interval? → draw a conclusion. You can only conclude whether the means differ; you cannot infer the source of the sample.

⑥ Factors affecting precision: increasing n ⇒ $\sigma_{\bar{X}}$ decreases, the interval becomes narrower, and the tail probability on one side far from μ becomes smaller; increasing the confidence level ⇒ the interval becomes wider; increasing σ (or s) ⇒ the interval becomes wider.